

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

26zfAn2dw6gNzb2yNVECGLBxgGxPG2qJdY3pKRTk2oX1

Date: 5/1/2025

Compounds: Aspirin, Famotidine, Laughing Gas

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Famotirin Nitroxylate

Formula: C18H23N3O5S

Weight: 393.46

Drug-likeness: 0.78

Activity:

Anti-inflammatory Analgesic with Gastric Protective Properties

Target Analysis

Cyclooxygenase-1 and -2 (COX-1 and COX-2)

Binding Affinity: -8.7

Inhibition reduces pain, inflammation and thromboxane-mediated platelet aggregation

Histamine H2 Receptor

Binding Affinity: -9.1

Blocking reduces gastric acid secretion

Nitric Oxide Synthase

Binding Affinity: -7.4

Mediates vasodilation and neurotransmission effects

Toxicity Profile

Risk Level: Moderate

Affected Systems: Gastrointestinal tract, Respiratory system, Central Nervous System

Mitigation Suggestions:

- Co-administration with gastroprotective drugs
- Avoid in patients with compromised respiratory function

Pharmacokinetics

Absorption:

Moderate oral bioavailability

Distribution:

Widespread with good CNS penetration

Metabolism:

Hepatic metabolism via cytochrome P450 enzymes

Excretion:

Primarily renal

Half-life: 5-8 hours

Detailed Analysis

Famotirin Nitroxylate represents a novel hybrid compound emerging from the integration of Aspirin, Famotidine, and Laughing Gas (nitrous oxide). By leveraging the analgesic, gastric protective, and vasodilatory properties respectively of its progenitors, this compound targets COX enzymes to provide pain relief, histamine H2 receptors to protect gastric mucosa, and utilizes nitric oxide-related pathways to instigate vasodilatory actions. Evaluated for drug-likeness and potential toxicity, Famotirin Nitroxylate presents a promising approach for addressing pain with minimized gastrointestinal disturbance and expanded therapeutic benefits. Analyzed computationally for binding affinities against key protein targets within relevant pathophysiological pathways, the compound exhibits competitive interactions suggesting efficacy in multi-faceted roles. Further investigational work is advised to optimize this potential therapeutic agent and mitigate adverse effects through rigorous testing and clinical trials.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>